

```

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload an image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Low Sampling (Reduce Resolution)
# -----

low_res = cv2.resize(
    img,
    (img.shape[1]//8, img.shape[0]//8),
    interpolation=cv2.INTER_AREA
)

low_res = cv2.resize(
    low_res,
    (img.shape[1], img.shape[0]),
    interpolation=cv2.INTER_NEAREST
)

# -----
# Improve Image Formation
# Sharpen Image
# -----

kernel = [[0,-1,0],
          [-1,5,-1],
          [0,-1,0]]

kernel = cv2.UMat.get(cv2.UMat(kernel)) if False else __import__("numpy").array(kernel)

sharp = cv2.filter2D(img, -1, kernel)

# -----
# Display
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_res)
plt.title("Low Resolution Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(sharp)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()

```

```
# -----  
# Explanation  
# -----  
  
print("Sampling:")  
print("- Low sampling reduces image resolution.")  
print("- Objects become blurry and difficult to detect.")  
  
print("\nImage Formation:")  
print("- Camera quality, focus, and lighting affect image clarity.")  
print("- Sharpening improves edge visibility.")  
  
print("\nApplications:")  
print("- Autonomous driving")  
print("- Surveillance")  
print("- Medical imaging")
```

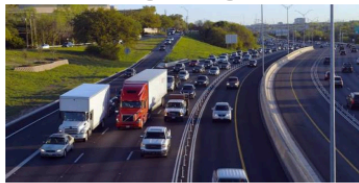
Upload an image

images.jpg

images.jpg(image/jpeg) - 36594 bytes, last modified: 14/07/2026 - 100% done

Saving images.jpg to images.jpg

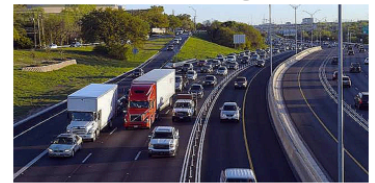
Original Image



Low Resolution Image



Enhanced Image



Sampling:

- Low sampling reduces image resolution.
- Objects become blurry and difficult to detect.

Image Formation:

- Camera quality, focus, and lighting affect image clarity.
- Sharpening improves edge visibility.

Applications:

- Autonomous driving
- Surveillance
- Medical imaging